

LAB 1: REQUIREMENTS ENGINEERING & UML USE-CASE MODELLING

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PROBLEM STATEMENT: #37

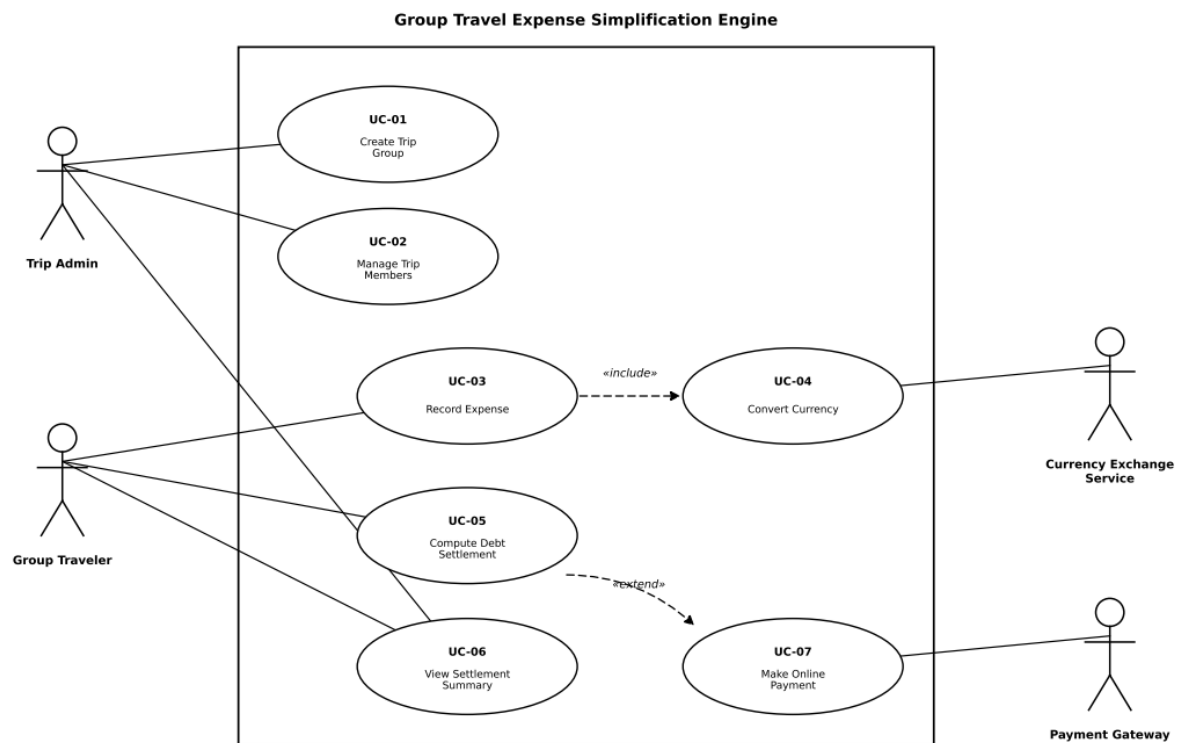
Retail, E-Commerce & Finance Group Travel Expense Simplification Engine

1. REQUIREMENTS TABLE:

Req. ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall compute the optimal debt settlement graph using min-flow algorithms to minimize total peer-to-peer payment transactions.	High	Pass: Group balance settles to net zero with minimal payment edges. Fail: Sum of debts does not equal sum of credits.	Reduces the number of repayments travellers must make and simplifies group settlement.
FR-002	Functional	The system shall allow a Group Traveler to record a shared expense by specifying the amount, currency, payer, and participating travellers.	High	Pass: Expense is saved with the correct amount, currency, payer, and participants. Fail: Expense cannot be saved when any required information is missing.	Accurate expense recording is required for calculating individual balances and settlement.
FR-003	Functional	The system shall automatically convert an expense recorded in a foreign currency into the trip's base currency using the applicable daily exchange rate.	High	Pass: Converted amount matches the original amount using the applicable daily exchange rate, within 0.01 rounding tolerance. Fail: An incorrect or unavailable rate is used.	Ensures expenses in different currencies can be compared and included correctly in settlement calculations.
FR-004	Functional	The system shall calculate and display the net balance of each Group Traveler based on all recorded group expenses.	High	Pass: The sum of all travellers' net balances equals zero. Fail: The calculated balances do not reconcile to zero.	Identifies how much each traveller owes or is entitled to receive before settlement.
FR-005	Functional	The system shall allow the Trip Admin to create a trip group, add or remove Group	Medium	Pass: The admin dashboard shows accurate per-traveller expense totals and	Gives the Trip Admin oversight of the group's shared finances — not just

		Tarvellers, to view the recorded group expenses, traveller balances, and calculated debt settlement transactions that is a consolidated dashboard for each.		balances matching each traveller's own view, and a removed member can no longer add expenses. Fail: The dashboard omits a member's activity, shows stale figures, or a non-admin can access it.	membership control — supporting accountability and dispute resolution across all travellers.
NFR-001	Nonfunctional	The settlement calculation engine must execute in under 100 ms for groups of up to 50 travellers with 500 expenses.	High	Pass: Benchmarking tests confirm that the settlement calculation completes in under 100 ms for 50 travellers and 500 expenses. Fail: Calculation takes 100 ms or longer.	Ensures fast settlement calculations even for relatively large groups.
NFR-002	Nonfunctional	The system shall maintain at least 99.5% monthly availability for expense recording and settlement services.	Medium	Pass: Availability monitoring shows at least 99.5% uptime over a rolling 30-day period. Fail: Monthly availability falls below 99.5%.	Travelers should be able to record expenses and calculate settlements reliably throughout a trip.

2. UML USE CASE DIAGRAM



User / Actor	Type	Role & Responsibilities
Group Traveler	Primary (Human)	Records shared expenses (amount, currency, participants), views their own balance and the group's settlement summary, and pays outstanding obligations via the payment gateway.
Trip Admin	Primary (Human)	Creates the trip group and sets its base currency, adds/removes Group Travelers, and has oversight access to a consolidated dashboard of every member's expenses and balances.
Currency Exchange Service	Secondary (External System)	Supplies the daily exchange rates the system uses to convert foreign-currency expenses into the trip's base currency.
Payment Gateway	Secondary (External System)	Processes and confirms online payments initiated by a Group Traveler to settle a debt.

3. KEY USE-CASE FLOW

Use Case: UC-05: Compute Debt Settlement

Primary Actor: Group Traveler

Secondary Actor: Trip Admin (may also trigger settlement review)

Related Requirement: FR-001, FR-002, FR-003

Preconditions

- The trip group exists and has at least two active members.
- At least one expense has been recorded and successfully converted to the trip's base currency.
- The requesting user is authenticated and is a member of the trip group.

Postconditions

- Success: A settlement graph is generated showing the minimum set of peer-to-peer payments such that total debts equal total credits (net balance = 0).
- Failure: No settlement graph is generated; existing balances remain unchanged and the error is logged for review.

Main Success Scenario

1. Group Traveler selects "Settle Up" for the trip.
2. System retrieves all recorded expenses for the trip and confirms each has been converted to the base currency.
3. System calculates each member's net balance (total paid minus total owed).
4. System runs a min-flow settlement algorithm to determine the minimum set of payments needed to zero out all balances.
5. System validates that the sum of all computed debts equals the sum of all computed credits.
6. System displays the settlement summary listing each pairwise payment obligation.
7. Group Traveler reviews the settlement summary.
8. Use case ends successfully.

Alternate Flow- 5a. *Balance Mismatch Detected*

- 5a1. System detects that the sum of debts does not equal the sum of credits (e.g., due to an unconverted or corrupted expense entry).
- 5a2. System displays a "Settlement Error" message and flags the offending expense(s) for review.
- 5a3. Group Traveler or Trip Admin corrects or removes the flagged expense.
- 5a4. System re-runs the settlement calculation, returning to step 3.